Enterprise Risk Management — Exercises & Case Studies

A mid-sized bank operates a trading desk managing a diversified market-risk portfolio with a total market value of **€100 million**. The bank has implemented an ERM framework centered on quantitative risk limits, with Value-at-Risk (VaR) as the primary risk metric.

How to use this student workbook

This student workbook contains a merged, pedagogical set of exercises on risk measurement (VaR and ES) and applied ERM case studies. Each exercise begins with a short contextual introduction that explains the learning objective. Where appropriate, you will be asked to work with a realistic, larger dataset; instructions for generating and analysing the data in **Excel** are provided. No solutions are included in this file.

Part I — Risk Measurement Foundations

1 Lecture 1: Intuition about VaR and ES

Learning objective

Understand what VaR and Expected Shortfall (ES) measure, their limitations, and how sample size and tail behavior affect them.

Exercise 1.1 — Conceptual VaR

A trading desk reports a **1-day VaR at 99% confidence** of **€10 million**.

1. In plain language, explain what this number means.
2. Over 100 trading days, how many days would you expect losses to exceed €10 million?
3. Does VaR tell you how large losses can be beyond €10 million? Explain.
4. Give two short examples of situations where VaR can be misleading.

Exercise 1.2 — Historical VaR with a realistic sample (Excel)

Context: Small samples (10 observations) are useful for demonstrating mechanics but not for building intuition about model stability. In this exercise you will create a realistic 250-day return series in Excel with occasional large negative shocks and compute historical VaR and ES.

Excel data generation (no code required):

1. Open a new Excel workbook. In column A (A2:A251) create 250 rows labelled Day 1 to Day 250.
2. In column B (B2:B251) generate daily returns using Excel's random functions and a simple volatility clustering approximation:
   - In B2 enter: $=\mathrm{NORM.INV(RAND(),0.0004,0.01)}$ to create a base normal return.
   - In C2 enter a volatility factor: $=1 + 0.5*\mathrm{ABS(NORM.INV(RAND(),0,0.8))}$.
   - In D2 compute the adjusted return: $=\mathrm{B2*C2}$.
   - Copy B2:D2 down to row 251.
3. To inject occasional negative jumps, choose five row indices (for example rows 32, 80, 122, 192, 232) and in column D for those rows subtract a jump value (e.g., -0.05, -0.08, or -0.12). Example: if D32 currently contains a value, replace it with $=\mathrm{D32}$ - 0.08.
4. Save the sheet as $\mathrm{returns.xlsx}$.

Excel calculations for VaR and ES:

1. In column E compute losses as negative returns: $=$ -$\mathrm{D2}$ and copy down.
2. Sort a copy of column E in descending order (largest losses first) using Excel's SORT or by copying to a new sheet and using Data → Sort.

3. 95% historical VaR: locate the 95th percentile of losses using $=\text{PERCENTILE.EXC(E2:E251,0.95)}$.
4. 99% historical VaR: $=\text{PERCENTILE.EXC(E2:E251,0.99)}$.
5. 95% ES: compute the average of losses greater than or equal to the 95% VaR using $=\text{AVERAGEIF(E2:E251,”>=”}$ $\text{cell}_w ith_V aR95)$.
5. 99% ES: $=\text{AVERAGEIF(E2:E251,”>=”}$ $\text{cell}_w ith_V aR99)$.
5. Convert to euros for a €50m portfolio: multiply the loss fractions by 50,000,000.

Deliverables: a short report (1 page) with the computed VaR/ES numbers (in), a histogram (Excel chart of column D), the backtest count (number of days where loss > VaR99), and a 3–4 sentence interpretation.

2 Lecture 2: Model risk and stress testing

Learning objective

Learn how model assumptions (normality, stationarity) and stress scenarios affect risk metrics and decision-making.

Exercise 2.1 — Parametric VaR vs Historical VaR (Excel)

1. Using the synthetic dataset in returns.xlsx, compute the sample standard deviation of returns with =STDEV.S(D2:D251).
2. Parametric 99% VaR (assuming normality) = =NORM.S.INV(0.99) * stdev, where stdev is the cell with the standard deviation.
3. Compare the parametric VaR to the historical VaR computed earlier and write a short explanation of the difference.
4. Stress test: compute the standard deviation of the most recent 20 days with =STDEV.S(D232:D251) and double it. Recompute parametric VaR using the doubled volatility and comment on the change.

Exercise 2.2 — Scenario analysis (Excel)

1. Construct a stress scenario for the €100m portfolio in Case 1: specify an equity shock (e.g., -8%), an interest-rate shock (e.g., +30 bps), and a liquidity shock (e.g., margin calls equal to X% of buffer). Use Excel to compute the P&L impact:
   - Equity P&L = 60,000,000 * equity shock.
   - Interest-rate P&L = sensitivity per bp * bp shock.
   - Liquidity impact: specify a margin outflow and compute required asset sales if buffer is exceeded.
2. Explain in a short paragraph how scenario analysis complements VaR and ES in ERM.

Part II — Applied ERM Case Studies

3 Case 1: Market risk, VaR limits and stress

Context (excerpt)

A mid-sized bank operates a trading desk managing a diversified market-risk portfolio with a total market value of **€100 million**. The bank has implemented an ERM framework centered on quantitative risk limits, with Value-at-Risk (VaR) as the primary risk metric.

Portfolio composition and parameters

- Equity exposure: 60m; daily equity volatility: 2%.
- Interest-rate exposure: 40m; daily yield volatility: 5 bps; sensitivity: +1 bp $\Rightarrow$ 0.20m loss.
- Correlation between equity returns and yield changes: 0.
- 1-day 99% VaR limit: 3.0m.

Exercise 3.1

1. Compute the 1-day 99% VaR of the portfolio using a variance–covariance approach (show intermediate steps in Excel).
2. Does the desk comply with the VaR limit? Explain.
3. Consider a stress scenario: equity decline 8%, yields +30 bps. Compute the scenario loss in Excel.
4. Explain why such a stress loss can occur even if VaR is within limits.
5. Propose two complementary ERM controls (non-VaR) that would reduce the chance of surprise losses.

4 Case 2: Expected Shortfall, governance and escalation

Context

The bank now monitors 1-day 99% Expected Shortfall (ES) with a limit of 4.0m and mandatory escalation on breach. The last 12 trading days' P&L (in m) are:

Day	P&L (m)	Day	P&L (m)
1	+0.6	7	-3.5
2	-1.1	8	+0.5
3	+0.4	9	-1.7
4	-2.3	10	-4.2
5	+0.9	11	-0.6
6	-0.8	12	-6.8

Exercise 3.2

1. Sort the P&L distribution from worst to best (Excel: use SORT or sort the column).
2. Compute the empirical 1-day 99% ES and determine whether the ES limit is breached (Excel: use $=\mathrm{AVERAGEIF}(\mathrm{range}, ">="\ \ \mathrm{VaR99})$ after computing VaR99 with $\mathrm{PERCENTILE.EXC}$).
3. The large loss on Day 12 occurred without prior escalation. Identify possible failures in: risk reporting, governance architecture, and incentives/risk culture.
4. Explain why replacing VaR with ES alone does not guarantee effective ERM.
5. Propose a practical escalation rule (quantitative + qualitative) that would have forced earlier action in this scenario.

5 Case 3: Signals without escalation (governance and culture)

Context

A bank expands rapidly into structured finance. Risk reports show rising concentration, frequent exceptions, and stress-test sensitivity. Committee minutes use soft language and leave follow-up to business judgment. A downturn causes large losses and supervisory scrutiny.

Exercise 3.3

1. Based on the description, identify the type of ERM framework (principles-based vs rules-based) and justify your answer.
2. Reconstruct the ERM architecture: who takes risk, who monitors, who decides escalation?
3. List the early warning signals and explain why they failed to trigger corrective action.
4. Explain how incentives and culture can undermine ERM, giving two concrete examples.
5. Propose two concrete governance changes that would make escalation automatic when certain signals appear.

6 Case 4: Hedging effectiveness (prospective vs retrospective)

Context

A firm hedges commodity exposure using futures. Normal-period estimates (24 months) and stress-period realized statistics are provided. The firm uses the minimum-variance hedge ratio and evaluates hedging effectiveness by variance reduction. Margining creates liquidity exposure.

Data (monthly, m)

	Normal period	Stress period
E[X]	0.0	-2.0
Var(X)	100	225
Var(H)	64	144
Cov(X,H)	48	12

Initial margin: 10m; average monthly variation margin outflow: 18m; liquidity buffer: 25m.

Exercise 3.4

1. Compute the minimum-variance hedge ratio h^* using normal-period statistics (Excel: compute $=\mathrm{Cov}/\mathrm{VarH}$).
2. Using h^*, compute prospective hedged variance and hedging effectiveness (HE$_{pros}$) in Excel.
3. Using the same h^* but stress-period statistics, compute retrospective hedged variance and HE$_{retro}$.
4. Explain why HE$_{retro}$ differs from HE$_{pros}$ in terms of correlation, basis risk, and regime change.
5. Is the liquidity buffer sufficient on average? What governance changes would reduce the probability of forced asset sales if margin spikes to 40m?
6. Identify three risk categories that interacted in this episode and explain how ERM should govern their interaction.

7 Case 5: When hedging increased risk

Context

A large industrial firm hedges commodity price risk with derivatives. Ex ante the hedge reduces variance by 75% (h* = 0.9). After a shock, correlations weaken, volatility spikes, and margin calls create liquidity stress. HE$_{retro} < 0$.

Exercise 3.5

1. Identify the main risks and how they interact.
2. Explain why the hedge looked effective ex ante.
3. Interpret the hedge ratio h^* and its assumptions.
4. Explain why prospective effectiveness failed to predict outcomes.
5. Which secondary risks were insufficiently considered?
6. Propose two ERM governance changes that would have reduced the crisis severity.

8 Case 6: Shell — ERM at a large multinational

Context (excerpt)

Shell operates across upstream, integrated gas, chemicals, trading, power, and retail. Its ERM is principles-based, scenario-driven, and integrates risk into strategy and capital allocation.

Exercise 3.6

Answer the following (short essay style, 200–400 words each):

1. Characterize Shell's ERM framework: rules-based or principles-based? Compliance-oriented or decision-oriented? Support your answer with evidence.
2. Why does Shell use scenario analysis for long-term risks? What are the limits of quantitative measurement?
3. How is risk ownership and oversight allocated? When could this separation weaken risk management?
4. Identify three principal risks and explain how Shell integrates them into strategy.
5. Describe a plausible failure mode for Shell's ERM and the early warning signals you would monitor.

Part III — Group ERM Design Exercise

Group assignment (included)

Learning objective

Design a realistic ERM system for a complex, capital-intensive multinational. The exercise tests synthesis: governance, risk identification, measurement, response, and integration into strategy.

Task

Prepare a 2–3 page ERM design proposal covering:

1. ERM philosophy and scope (principles vs rules; focused vs comprehensive).
2. ERM architecture and governance (who owns risks, who oversees, escalation).
3. Risk identification and assessment (how to capture emerging and interacting risks).
4. Risk response and instruments (how to govern derivatives, insurance, contingent liquidity).
5. Integration into strategy and capital allocation.
6. Reporting, dashboards, and escalation rules.
7. Two plausible failure modes and early warning signals.

Deliverable

A 2–3 page written proposal and a 10-minute group presentation. Include a one-page dashboard mock-up (PDF or slide) showing key indicators and escalation triggers.

End of student workbook.